

Network Trace Analysis Report: ARP Spoofing Capture

Overview

This packet capture documents activity on a small local network where an ARP spoofing attack is taking place. The trace contains ARP exchanges along with normal IP traffic between hosts. By examining the ARP behavior and packet flow, it is possible to identify the gateway, attacker, victim, and confirm whether the spoofing attempt succeeds.

Network Roles Observed

Early ARP traffic shows multiple hosts attempting to resolve the same IP address. i.e. 192.168.1.1. This pattern is consistent with gateway behavior as hosts must resolve the router's MAC address to reach external networks. One MAC address responds consistently as the legitimate owner of this IP, establishing the normal gateway mapping.

But later in the trace, a different host repeatedly sends unsolicited ARP replies claiming that the gateway IP belongs to its own MAC address. These replies are not triggered by requests and appear periodically, which is characteristic of ARP poisoning attempts.

Evidence of ARP Spoofing

ARP spoofing relies on sending forged ARP replies to overwrite a victim's ARP cache. In this capture:

- A host continuously advertises that 192.168.1.1 maps to its own MAC address.
- These forged replies target another host that is actively communicating with the gateway.
- The repeated nature of these announcements suggests an attempt to maintain false ARP cache entries.

This behavior is a clear indicator of an ARP spoofing attack rather than normal ARP resolution.

Victim Identification

The victim host can be identified by observing which device continues to exchange IP packets after receiving spoofed ARP responses. Its communication pattern shows that traffic intended for the gateway is redirected toward the attacker's MAC address.

This confirms that the victim's ARP cache has been poisoned, causing it to forward traffic through the attacker.

Traffic Flow after Poisoning

Following the spoofing activity, IP packets continue flowing between the victim and gateway. However, the Ethernet frames reveal that traffic is now routed through the attacker's interface. This indicates a successful man-in-the-middle position:

- Victim → attacker → gateway
- Gateway → attacker → victim

Such positioning allows the attacker to observe or modify packets without interrupting communication.

Capture Perspective

The trace contains packets exchanged among multiple hosts, not just traffic involving a single endpoint. This suggests the capture point had broad visibility of the network, consistent with monitoring from a mirrored switch port or a similar observation point.

Conclusion

The packet capture clearly demonstrates a successful ARP spoofing attack. One host impersonates the gateway by sending forged ARP replies, poisoning a victim's ARP cache. Subsequent traffic confirms that the attacker is positioned between the victim and the gateway, enabling interception of network communication.

This trace provides a practical example of how ARP spoofing operates, how it appears in packet captures, and how attackers can redirect traffic without disrupting connectivity.